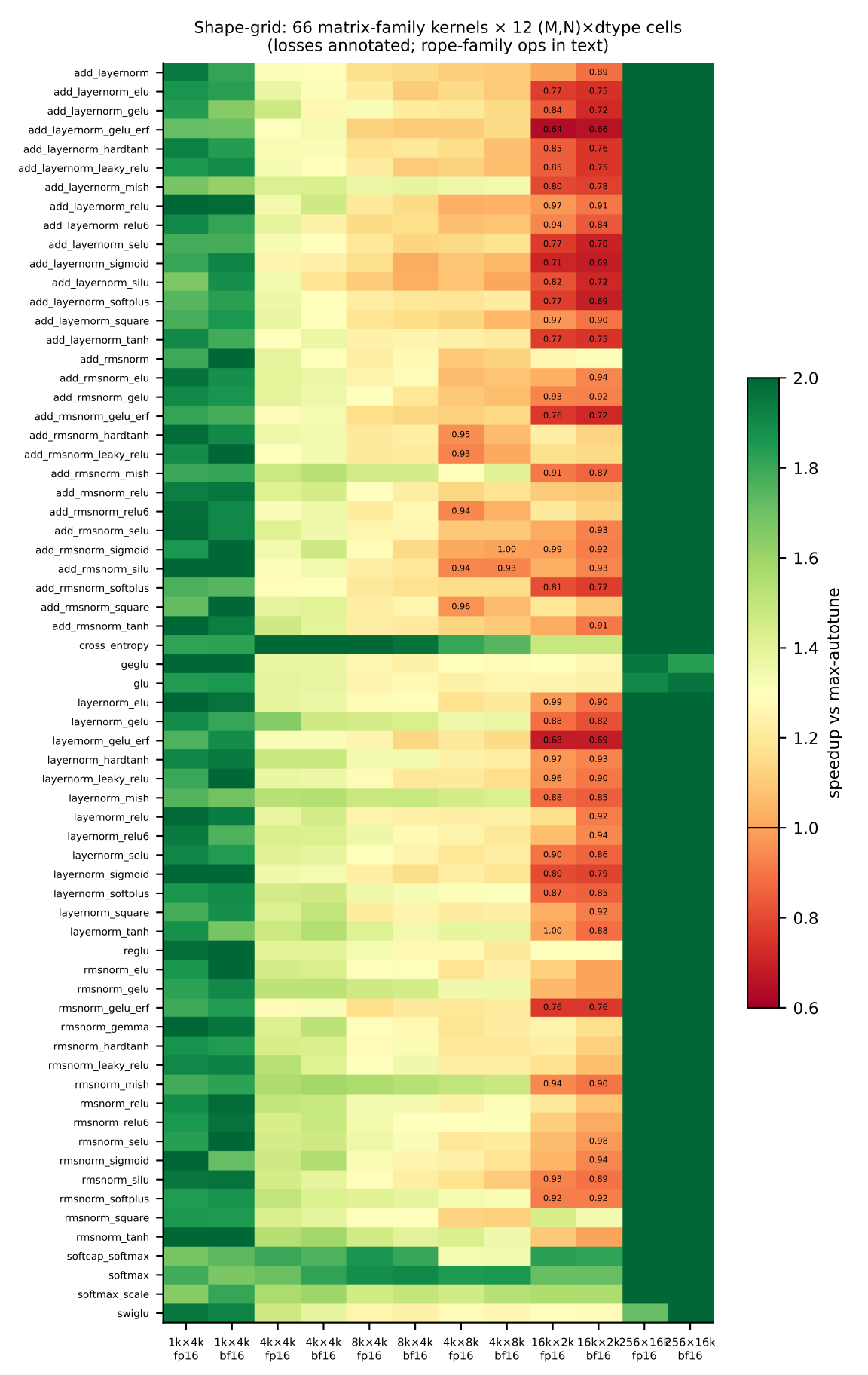
Shape-grid: 66 matrix-family kernels × 12 (M,N)×dtype cells
(losses annotated; rope-family ops in text)
add_layernorm
add_layernorm_elu
add_layernorm_gelu
add_layernorm_gelu_erf
add_layernorm_hardtanh
add_layernorm_leaky_relu
add_layernorm_mish
add_layernorm_relu
add_layernorm_relu6
add_layernorm_selu
add_layernorm_sigmoid
add_layernorm_silu
add_layernorm_softplus
add_layernorm_square
add_layernorm_tanh
add_rmsnorm
add_rmsnorm_elu
add_rmsnorm_gelu
add_rmsnorm_gelu_erf
add_rmsnorm_hardtanh
add_rmsnorm_leaky_relu
add_rmsnorm_mish
add_rmsnorm_relu
add_rmsnorm_relu6
add_rmsnorm_selu
add_rmsnorm_sigmoid
add_rmsnorm_silu
add_rmsnorm_softplus
add_rmsnorm_square
add_rmsnorm_tanh
cross_entropy
geglu
glu
layernorm_elu
layernorm_gelu
layernorm_gelu_erf
layernorm_hardtanh
layernorm_leaky_relu
layernorm_mish
layernorm_relu
layernorm_relu6
layernorm_selu
layernorm_sigmoid
layernorm_softplus
layernorm_square
layernorm_tanh
reglu
rmsnorm_elu
rmsnorm_gelu
rmsnorm_gelu_erf
rmsnorm_gemma
rmsnorm_hardtanh
rmsnorm_leaky_relu
rmsnorm_mish
rmsnorm_relu
rmsnorm_relu6
rmsnorm_selu
rmsnorm_sigmoid
rmsnorm_silu
rmsnorm_softplus
rmsnorm_square
rmsnorm_tanh
softcap_softmax
softmax
softmax_scale
swiglu
1k×4k fp16
1k×4k bf16
4k×4k fp16
4k×4k bf16
8k×4k fp16
8k×4k bf16
4k×8k fp16
4k×8k bf16
16k×2k fp16
16k×2k bf16
256×16k fp16
256×16k bf16
speedup vs max-autotune
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6